

Personalized Neural Distortion Models Recover Subject-Specific Color Vision Deficiency Profiles

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Abstract

Color vision deficiency (CVD) is typically characterized by categorical color confusion, yet the functionally consequential deficit may lie in the continuous interpolation of intermediate hues. We fit parsimonious, retina-inspired forward models to fMRI-measured hue interpolation profiles in area hV4 of two CVD participants and one normal control. Each CVD participant is best captured by a *different* model—a 1-parameter cone-shift for moderate protanomaly and a 2-parameter retinal–cortical model for mild deuteranomaly—while the normal control is correctly rejected by all models. The fitted models further reveal that correction feasibility is severity-dependent: the mild case admits exact stimulus-space inversion, whereas the moderate case produces irreversible opponent-arc compression under the best-fitting retinal model. These results establish a proof-of-concept framework for individualized neural distortion assessment in CVD, with behavioral validation as the critical next step.

1. Introduction

Color vision deficiency (CVD) arises from altered spectral sensitivity of retinal cone photoreceptors and affects approximately 8% of males worldwide (Machado et al., 2009). Current assistive approaches, including commercial spectral-notch filters, apply a fixed, generic correction. Recent evidence indicates that such filters may alter color *appearance* without improving color *discrimination* (Somers & Others, 2024), suggesting that effective correction may require individually-tailored strategies.

A key reason generic approaches may be insufficient is that CVD-related neural distortion is not uniform across

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individuals. Even within a single diagnostic category (e.g., deuteranomaly), the degree of cortical compensation varies substantially (Tregillus et al., 2021), and the functional signature that matters for continuous color tasks—the ability to interpolate between adjacent hues, not merely classify them—may differ in ways that categorical diagnosis cannot capture.

Here we propose an individualized approach: fitting parsimonious mechanistic forward models to each CVD participant’s neural interpolation profile, as measured by leave-one-color-out (LOCO) cross-validation in fMRI. Our contributions are:

1. We show that CVD is better characterized by *subject-specific interpolation failure* than by coarse categorical deficits, and that this failure is localized to area hV4.
2. We introduce a personalized distortion-fitting framework in which different CVD participants are best captured by different parsimonious models (1–2 DOF), with successful specificity rejection of a normal control.
3. We demonstrate that correction feasibility is severity-dependent: the fitted forward model determines whether exact stimulus-space correction is achievable or structurally precluded.

2. Methods

2.1. Participants and Stimuli

Ten participants were included in the analysis (7 healthy controls [HC], 3 females, age 22.7 ± 2.5 years; 3 color vision deficient [CVD], all male, including one deuteranope, one protanomalous, and one near-normal deutan serving as a specificity control). Color vision was assessed using the 14-plate Ishihara test (Ishihara, 1917). Given the small CVD sample, all CVD results are interpreted as individual case demonstrations, not population-level effects (Crawford & Howell, 1998).

Stimuli consisted of eight colors equally spaced from 0° to 315° in CIE $L^*a^*b^*$ space, with lightness fixed at $L^* = 75$

and chroma distance of 40 from the achromatic point. Participants performed a modified Rapid Serial Visual Presentation (RSVP) detection task (Brouwer & Heeger, 2009) across six runs per session, maintaining fixation while viewing 1.5 s color patches presented in optimized order. Functional images were acquired on a Siemens 3T scanner (TR = 1.5 s, 2 mm isotropic voxels, 24 oblique slices perpendicular to the calcarine sulcus), preprocessed with fMRIPrep, and projected to retinotopically-defined area hV4 (Wang atlas, 50% threshold; Wang et al. 2015). Response amplitudes were estimated using a two-stage FIR-then-GLM procedure (Brouwer & Heeger, 2009) and Procrustes-aligned across runs within each participant to improve cross-run consistency.

2.2. Neural Phenotype: Interpolation Vulnerability

We modeled hue-selective neural populations using half-wave rectified and squared sinusoidal basis functions (Brouwer & Heeger, 2009): each of $K=3$ hypothetical channels was tuned to a preferred hue θ_k , with response $r_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$. The weight matrix W mapping channel outputs to voxel responses was estimated via ridge regression with generalized cross-validation (GCV) for regularization (Golub et al., 1979).

The target phenotype is the *LOCO vulnerability profile*. Leave-one-color-out (LOCO) cross-validation tests whether the model can interpolate to novel hues not seen during training, assessing the preservation of continuous color-space geometry: W is trained on 7 of 8 colors and voxel responses to the held-out color are predicted. The per-color prediction error defines an 8-element vulnerability vector $\mathbf{v} \in \mathbb{R}^8$ for each participant, capturing which hues are poorly interpolated from their neighbors.

This phenotype is specific to continuous interpolation and distinct from color discrimination. Leave-one-run-out (LORO) cross-validation, which tests whether individual colors can be reliably distinguished across repetitions, shows no HC-CVD difference ($p = 0.668$), while LOCO interpolation error is significantly elevated in CVD specifically in hV4 ($p = 0.017$).

2.3. Distortion Models

We fit three forward models of hue distortion (Figure 1a), each parameterizing how a physical stimulus angle θ is transformed into a perceived angle θ' :

Machado cone shift (1 DOF). A physiological model (Machado et al., 2009) in which the anomalous cone’s spectral peak is shifted by $\Delta\lambda \in [0, 20]$ nm. This retinal-level model predicts hue distortion through altered L–M cone opponency.

Retinal + cortical gain (R+C; 2 DOF). Extends the cone shift with a cortical opponent-channel gain g :

$$rg' = rg_{\text{base}} + (1+g) \cdot (rg_{\text{ret}} - rg_{\text{base}}) \quad (1)$$

where $g=0$ reduces to pure Machado, $g=-1$ is exact cortical compensation, and $g<-1$ is overcompensation (Tregillus et al., 2021).

Fourier warp (4 DOF; overfitting ceiling). A model-free sinusoidal distortion field: $\delta(\theta) = a_1 \sin \theta + b_1 \cos \theta + a_2 \sin 2\theta + b_2 \cos 2\theta$. With 4 parameters for 8 data points, this serves as a *ceiling* to assess whether parsimonious models leave substantial variance unexplained.

2.4. Fitting and Specificity

For each model and candidate parameter(s), we simulate the LOCO vulnerability profile that a healthy observer would exhibit if viewing stimuli distorted by the model’s forward map. The fit is the Spearman rank correlation ρ between simulated and observed vulnerability vectors, optimized over a grid ($\Delta\lambda$: 41 points; g : 25 points; Fourier: differential evolution).

Statistical significance is assessed by exact permutation of the 8-color labels ($8! = 40,320$ permutations). A valid model must satisfy two criteria:

1. **Sensitivity:** significant fit ($p < 0.05$) for CVD participants.
2. **Specificity:** non-significant fit for the normal control (sub-10).

A model that fits CVD subjects but also yields a false positive on the control is rejected as overfitting.

3. Results

3.1. Subject-Specific Model Selection

Table 1 and Figure 1 present the central result. Each CVD participant is best captured by a different parsimonious model, and the normal control is correctly rejected.

Sub-09 (moderate protanomaly). The 1-DOF Machado model with $\Delta\lambda = 13.5$ nm—within the established range for moderate–severe protanomaly (Machado et al., 2009)—captures the interpolation profile at $\rho = 0.762$, $p = 0.018$. Adding a cortical gain parameter does not improve fit ($g = 0.0$ at optimum), indicating that a purely retinal account is sufficient.

Sub-08 (mild deuteranomaly). The 1-DOF Machado model is only trend-level ($p = 0.058$). The 2-DOF R+C model, with a small retinal shift ($\Delta\lambda = 2.0$ nm) and a large

Table 1. Subject-specific distortion model fits (hV4 LOCO). Each CVD subject is best explained by a different parsimonious model. The normal control is correctly rejected by all models. Fourier serves as an overfitting ceiling.

Subject	Best model	DOF	ρ	p_{perm}
Sub-08 (deutan)	R+C	2	0.857	0.005**
Sub-09 (protan)	Machado	1	0.762	0.018*
Sub-10 (control)	—	—	-0.048	0.559
<i>Overfitting ceiling (Fourier, 4 DOF):</i>				
Sub-08	Fourier	4	0.976	0.0002
Sub-09	Fourier	4	0.762	0.018

[FIGURE PLACEHOLDER]

Figure 1: Subject-Specific Distortion Profiles

(a) Model schematic: stimulus $\theta \rightarrow$ retinal cone shift ($\Delta\lambda$) $\rightarrow$ optional cortical gain (g) $\rightarrow$ distorted $\theta' \rightarrow$ forward encoding $\rightarrow$ LOCO evaluation.

(b–d) 8-color LOCO vulnerability profiles. Gray bars: observed CVD profile. Colored line: best-model fit. (b) Sub-08: R+C fit ($\rho=0.857$, $p=0.005$). (c) Sub-09: Machado fit ($\rho=0.762$, $p=0.018$). (d) Sub-10: flat profile, $p=0.559$ (specificity pass).

Figure 1. The central result. Each CVD participant exhibits a distinct vulnerability profile (gray bars) captured by a different parsimonious model (colored lines). The normal control shows no structured vulnerability. Panel (a) illustrates the model hierarchy.

cortical amplification ($g = +2.25$), achieves $\rho = 0.857$, $p = 0.005$. The cortical component is essential: the retinal shift alone explains little of the distortion.

Sub-10 (normal control). All parsimonious models yield $\Delta\lambda \approx 0$ and non-significant fits ($p = 0.559$), confirming specificity.

Model parsimony matters. The 4-DOF Fourier model achieves $\rho = 0.976$ for sub-08 but affords minimal constraint (4 free parameters for 8 colors). Its fit for sub-09 ($\rho = 0.762$) is identical to the 1-DOF Machado, demonstrating that extra flexibility yields no gain when the signal is well captured by a parsimonious model.

3.2. Fitted Parameters and Biological Plausibility

The fitted retinal parameters are physiologically plausible: sub-08’s $\Delta\lambda = 2.0$ nm matches very mild deuteranomaly, and sub-09’s $\Delta\lambda = 13.5$ nm matches moderate protanomaly (Machado et al., 2009). Sub-09’s cortical gain collapses to $g = 0$, consistent with a purely retinal account. Sub-08’s cortical gain ($g = +2.25$) indicates that hV4 *amplifies* rather than compensates the retinal deficiency—a novel finding without direct literature precedent, though cortical

[FIGURE PLACEHOLDER]

Figure 2: Severity-Dependent Correction Feasibility

(a) Sub-08 (mild): forward map is wide-arc; pre-image inversion restores all 8 colors (inner ring = original, outer = perceived after correction; dots overlap = success).

(b) Sub-09 (moderate): forward map compresses $360^\circ \rightarrow \sim 96^\circ$; three colors merge at $\sim 282^\circ$. Exact inversion impossible. Separation-optimized fallback: $5.6\times$ improvement, 48% of healthy separation.

Figure 2. The forward model determines correction limits. Mild CVD admits exact stimulus-space restoration; moderate CVD under the best-fitting retinal model does not.

compensation magnitudes vary widely across individuals (Tregillus et al., 2021).

3.3. Correction Feasibility Is Severity-Dependent

Given a fitted forward model $D: \theta \rightarrow \theta'$, we ask whether an inverse correction exists—i.e., whether modified stimulus angles θ^* can be found such that $D(\theta^*) = \theta_{\text{target}}$ for all 8 colors.

Sub-08 (mild): The R+C forward map preserves an opponent arc wide enough that 8/8 colors remain distinguishable. Numerical pre-image search yields exact solutions ($< 0.001^\circ$ residual), producing an 8-point stimulus correction lookup table.

Sub-09 (moderate): The Machado forward map with $\Delta\lambda = 13.5$ nm compresses the 360° opponent arc to approximately 96° , merging green, cyan, and blue into a single perceived angle ($\sim 282^\circ$). Exact inversion is structurally impossible for 4/8 colors (Figure 2). A discriminability-oriented fallback (maximizing minimum pairwise angular separation) achieves $5.6\times$ improvement over the unfiltered state but reaches only 48% of the healthy-observer separation.

This severity-dependent feasibility boundary is the forward model’s *prescriptive* contribution: it does not merely characterize distortion but predicts *a priori* whether stimulus-space correction can succeed for a given individual.

4. Discussion

What this framework does and does not establish. We have shown that subject-specific interpolation profiles in CVD can be recovered by parsimonious forward models, with correct specificity rejection and physiologically plausible parameters. We have *not* shown that the resulting cor-

rections improve perceptual discrimination—that requires behavioral validation, which is future work. The current evidence supports the claim that the method identifies individual distortion profiles and estimates whether stimulus-space correction is feasible under the fitted model; it does not yet support the stronger claim that these corrections restore perception.

Why different models for different subjects? Sub-09’s profile is captured by pure cone shift ($g = 0$) while sub-08 requires a cortical component ($g = +2.25$). One interpretation is that sub-08’s mild retinal deficit ($\Delta\lambda = 2.0$ nm) produces a small signal that cortical gain amplifies into a measurable interpolation deficit—an effect invisible under retinal-only modeling. More conservatively, g may be absorbing non-retinal variance (e.g., attention, voxel noise) rather than reflecting a genuine cortical process. Disentangling these accounts requires independent physiological measures.

Why reject more flexible models? The Fourier model ($\rho = 0.976$ for sub-08) fits nearly perfectly but affords poor generalization guarantees with 4 parameters for 8 data points. The critical test is not fit quality but the joint requirement of sensitivity and specificity: a valid model must explain CVD profiles *while rejecting the normal control*. Under this criterion, parsimonious models are the meaningful result.

Limitations. Three CVD participants are insufficient for population-level claims. This work is a proof-of-concept for individualized neural distortion modeling, not an epidemiological study. The fMRI requirement limits near-term scalability. The interpolation phenotype depends on forward encoding model specification (basis type, regularization), though the LOCO metric has been validated as robust across reasonable choices in prior analyses.

Relation to prior work. Our approach extends Brouwer & Heeger’s (Brouwer & Heeger, 2009) forward encoding framework—which demonstrated LOCO reconstruction in hV4 for healthy observers—to CVD with individual-level distortion fitting. The distinction between preserved classification and impaired interpolation is consistent with reports that CVD individuals maintain categorical but not continuous color representations (Bannert & Bartels, 2018). The cortical compensation parameters relate to accumulating evidence for long-term neural adaptation in CVD (Tregillus et al., 2021; Emery et al., 2021).

5. Conclusion

We present a proof-of-concept framework for individualized neural distortion modeling in color vision deficiency.

The framework identifies subject-specific hue interpolation failures, fits them with parsimonious mechanistic models, and predicts whether stimulus-space correction is feasible for a given individual. The critical next step is behavioral validation: determining whether the model-prescribed corrections actually improve color discrimination in the fitted individuals.

Impact Statement

This work develops personalized neural models of color vision deficiency. Potential positive impacts include individually-tailored color corrections for digital displays and accessibility tools. The framework currently requires fMRI, limiting immediate clinical deployment. Behavioral validation and informed consent are essential prerequisites before any correction is applied, as a misspecified model could impair rather than improve visual function.

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A. Supporting Analyses

A.1. Cross-Criterion Validation: Δ RDM Geometry

Beyond the primary LOCO fitting (main text), we performed an independent geometric validation using pairwise representational distance matrices (RDMs) in V1. A 2-component angular dilation model—parameterized by S-cone expansion β_s and confusion-axis modulation β_c —was fitted to the voxel-space $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$. Sub-09 achieved crossnobis cosine $p = 0.007$ with $\beta_s = 23.0^\circ \pm 10.2^\circ$ (bootstrap CI excluding 0). The cross-subject mean $\beta_s \approx 21.5^\circ$ matches the 21.4° B-Y axis rotation independently reported from behavioral hue-scaling (Emery et al., 2021)—convergent evidence from an entirely different fitting criterion and ROI.

This supplementary analysis confirms that the distortion captured by hV4 LOCO is not criterion-specific: it is also visible as geometric distortion in V1 representational structure.

A.2. Model Comparison: Full Table

Table 2. Full model comparison across hV4 LOCO and V1 Δ RDM criteria.

Subject	Model	DOF	hV4 LOCO ρ	hV4 LOCO p	V1 Δ RDM	Notes
Sub-08	Machado	1	0.619	0.058	anti-corr.	Trending only
	R+C	2	0.857	0.005**	0.179	Primary
	2-Comp	2	0.881	0.004**	CI excl. 0	Alternative
	Fourier	4	0.976	0.0002	—	Ceiling
Sub-09	Machado	1	0.762	0.018*	+0.091	Primary
	R+C	2	0.762	0.018	0.026*	$g=0$ (= Machado)
	2-Comp	2	0.690	0.035*	0.007***	Cross-criterion
	Fourier	4	0.762	0.018	—	No gain
Sub-10	All	—	−0.048	0.559	—	Specificity pass

A.3. Correction Feasibility Details

Sub-08 (R+C pre-image). All 8 colors admit exact pre-image solutions (mean $|\delta| = 23.5^\circ$, max = 42.9° , residual $< 0.001^\circ$). The resulting 8-point lookup table is directly implementable on any digital display.

Sub-09 (Machado pre-image). At $\Delta\lambda = 13.5$ nm, the opponent-space arc compresses from 360° to $\sim 96^\circ$. Colors c4 (green), c5 (cyan), and c6 (blue) converge to a single perceived angle ($\sim 282^\circ$). Exact pre-image: 4/8 colors. Separation-optimized fallback: min separation $1.03^\circ \rightarrow 5.76^\circ$ ($5.6\times$), mean separation $39.3^\circ \rightarrow 24.3^\circ$ (healthy: 70.7°).